



UK Government

 TAG International

Africa
Cyber
Programme

CYBER INVESTIGATIONS

Open-Source Intelligence | Dark Web | Cryptocurrency

4-Day Course (15 Sep - 18 Sep 2026)

PARTNERING TOWARDS A GLOBAL
CYBER VISION

DAY 3

Infrastructure, OSINT and Threat Actor Development

CRYPTOCURRENCY FUNDAMENTALS

What is Cryptocurrency?

-  **Digital Asset**
Exists electronically rather than as physical notes or coins
-  **Decentralised**
Operates without a central bank or controlling authority
-  **Cryptography**
Secure transactions and control access
-  **Scale**
Estimated 17,000 different cryptocurrencies globally



Blockchain is the underlying technology that records cryptocurrency transactions

What is Blockchain?



Distributed Ledger

A shared digital record of transactions



Immutable

Once confirmed & recorded, transactions cannot be changed or deleted



Blocks

Transactions are grouped together into blocks



Linked Records

Each block is cryptographically linked to the previous block

Transaction Hash	Method	Block	Age	From	To	Amount	Txn Fee
0x66017fdbf22...	Transfer	22078211	13 secs ago	titanbuilder.eth	Lido: Execution Layer R...	0.021062025 ETH	0.00001023
0x73cb812716...	Commit Blob	22078211	13 secs ago	Base: Batch Sender	Null: 0xff00...453	0 ETH	0.00011492
0xdf27f4d5d6c...	Propose Bloc...	22078211	13 secs ago	0x9a5Cc6E3...c80139C09	Taiko: L1	0 ETH	0.00065289
0x687116e39c...	Transfer	22078211	13 secs ago	Rollbit: ERC-20 Hot W...	0xa2f58fBD...D571625Da	0.002150217 ETH	0.00000973
0xc003a4a678f...	Withdraw	22078211	13 secs ago	punksotc.eth	CryptoPunks: C Token	0 ETH	0.00001424
0xbda85f55aee...	Transfer	22078211	13 secs ago	Bittrue	Proton Chain: XPR Tok...	0 ETH	0.00001603

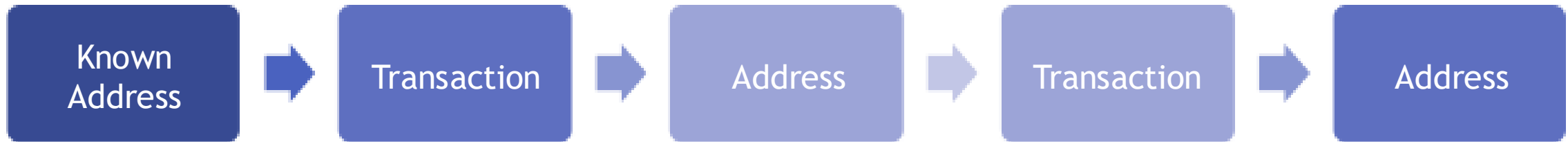
Crypto investigations can be complex - we are focussing on the basics

Investigative Requirement

- Cryptocurrency is used in ransomware, fraud, extortion and criminal activity
- Transactions recorded on a blockchain can provide a historical financial trail
- A cryptocurrency address can provide a start point for tracing funds
- Fund flows can lead to services such as cryptocurrency exchanges



Tracing Funds in Investigations



Follow the transactions

Each transaction may identify another address and provide the next investigative pivot



Corroborate attribution

A wallet can provide useful intelligence about a possible owner or service



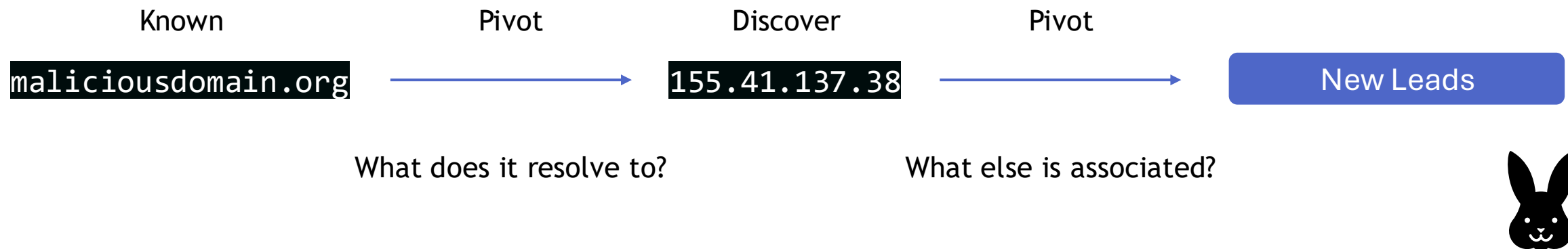
Look for patterns

Funds may move through several addresses, split apart or come together

Crypto investigations can be complex - we are only covering the basics

Pivoting the Investigation

Investigations develop by using what is already known to identify what can be investigated next



Each pivot should answer an investigative question

Historical Analysis

- Archive of historical versions of websites
- Shows how a site has changed over time
- Images and graphics may also be archived
- Can reveal investigative leads no longer visible on the live site
- Can connect current activity to previous campaigns or infrastructure



LIMITATION: Onion/dark web services are generally not archived by the Wayback Machine

WHOIS Analysis

- An easily accessible record
- Sometimes reveals domain registrant - who owns the domain
- Sometimes reveals domain registrar - who registered the domain
- Often privacy protected to hide details

```
Domain name:
    google.co.uk

Data validation:
    Nominet was able to match the registrant's

Registrar:
    Markmonitor Inc. [Tag = MARKMONITOR]
    URL: https://www.markmonitor.com

Relevant dates:
    Registered on: 14-Feb-1999
    Expiry date: 14-Feb-2025
    Last updated: 13-Jan-2024

Registration status:
    Registered until expiry date.

Name servers:
    ns1.google.com
    ns2.google.com
    ns3.google.com
    ns4.google.com

WHOIS lookup made at 23:40:59 15-Apr-2024
```

Exercise Equivalents

WHOIS

INTERNET ARCHIVE
WayBackMachine

chaintrace.cc



Cryptocurrency Tracing

whoisfinder.org



Domain Analysis

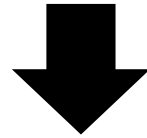
pagevault.cc



Historical Webpage Analysis

Evidence Location

Stage 3 & 4
Artefacts



<https://OperationGreyharbourTraining/greyharbour-trainee-pack>

Record your investigation using your CSV records